DFIA299: Capstone Project in Digital Forensics Investigations

Lab Title: **The Grand Monarch Hotel Case**

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Course Name: **Capstone Project in Digital Forensics**

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Introduction:

The Grand Monarch Hotel case took place on May 20, 2025, leading to the untimely demise of Ms. Clara Benson in room number 718. The USB drive seized from Marcus Langston, who is the prime suspect in the murder, was sent for a digital forensics analysis. This document will outline the comprehensive forensic analysis that was carried out on the disk image Mrs_Clara_Benson_case_image.001.

Background: Clara Benson died under mysterious circumstances. There were initial clues that suggested Marcus Langston's involvement; he even made a partial confession. But more information regarding the conspiracy, along with any possible other culprits involved, was buried deep inside electronic media contained in the USB drive. The task at hand was to retrieve everything that was hidden, whether it was deleted files, secret codes, steganographic messages, or audio steganography.

Step 1: Image Verification (Integrity Check)

```
Image Verification Results:
 Verification started:  Sat May 24 16:50:37 2025
 Verification finished: Sat May 24 16:50:37 2025
 MD5 checksum:    583bb23631d54dfe5446f1f094681f30 : verified
 SHA1 checksum:   467176bdfffef80f8fde152a187575735023dbea : verified

┌──(atul㉿kali)-[~/campstoneproject]
└─$ md5sum Mrs_Clara_Benson_case_image.001
583bb23631d54dfe5446f1f094681f30  Mrs_Clara_Benson_case_image.001
```

Step 2: Disk Structure Analysis

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ mmls Mrs_Clara_Benson_case_image.001
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Primary Table (#0)
001:  -------   0000000000   0000000127   0000000128   Unallocated
002:  000:000   0000000128   0000198783   0000198656   NTFS / exFAT (0x07)
003:  -------   0000198784   0000204799   0000006016   Unallocated
```

Step3: Files List

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ fls -o 128 Mrs_Clara_Benson_case_image.001
r/r 4-128-1:     $AttrDef
r/r 8-128-2:     $BadClus
r/r 8-128-1:     $BadClus:$Bad
r/r 6-128-4:     $Bitmap
r/r 7-128-1:     $Boot
d/d 11-144-4:    $Extend
r/r 2-128-1:     $LogFile
r/r 0-128-6:     $MFT
r/r 1-128-1:     $MFTMirr
d/d 43-144-1:    $RECYCLE.BIN
r/r 9-128-8:     $Secure:$SDS
r/r 9-144-11:    $Secure:$SDH
r/r 9-144-14:    $Secure:$SII
r/r 10-128-1:    $UpCase
r/r 10-128-4:    $UpCase:$Info
r/r 3-128-3:     $Volume
r/r 38-128-1:    .dropbox.device
d/d 36-144-1:    System Volume Information
V/V 256:         $OrphanFiles
```

Step4: Recursive Search

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ fls -r -o 128 Mrs_Clara_Benson_case_image.001
r/r 4-128-1:    $AttrDef
r/r 8-128-2:    $BadClus
r/r 8-128-1:    $BadClus:$Bad
r/r 6-128-4:    $Bitmap
r/r 7-128-1:    $Boot
d/d 11-144-4:   $Extend
+ d/d 29-144-2: $Deleted
+ r/r 25-144-2: $ObjId:$O
+ r/r 24-144-3: $Quota:$O
+ r/r 24-144-2: $Quota:$Q
+ r/r 26-144-2: $Reparse:$R
+ d/d 27-144-2: $RmMetadata
++ r/r 28-128-4:        $Repair
++ r/r 28-128-2:        $Repair:$Config
++ d/d 31-144-2:        $Txf
++ d/d 30-144-2:        $TxfLog
+++ r/r 32-128-2:       $Tops
+++ r/r 32-128-4:       $Tops:$T
+++ r/r 33-128-1:       $TxfLog.blf
+++ r/r 34-128-1:       $TxfLogContainer00000000000000000001
+++ r/r 35-128-1:       $TxfLogContainer00000000000000000002
r/r 2-128-1:    $LogFile
r/r 0-128-6:    $MFT
r/r 1-128-1:    $MFTMirr
d/d 43-144-1:   $RECYCLE.BIN
+ d/d 44-144-5: S-1-5-21-3935823118-2491923564-2231019596-1001
++ r/r 50-128-1:        $I5VUNDH.docx
++ r/r 48-128-1:        $I8BDGE1.jpg
++ r/r 53-128-1:        $IQTQ5UY.docx
++ r/r 52-128-1:        $IVMQAU9.txt
++ r/r 51-128-1:        $IYT9YUG.txt
++ r/r 47-128-1:        $IYW9K6B.bmp
++ r/r 42-128-1:        $R5VUNDH.docx
++ r/r 41-128-1:        $R8BDGE1.jpg
++ r/r 49-128-3:        $RQTQ5UY.docx
++ r/r 40-128-1:        $RVMQAU9.txt
++ r/r 39-128-1:        $RYT9YUG.txt
++ r/r 46-128-1:        $RYW9K6B.bmp
++ r/r 45-128-1:        desktop.ini
r/r 9-128-8:    $Secure:$SDS
```

```
+ r/r 24-144-2: $Quota:$Q
+ r/r 26-144-2: $Reparse:$R
+ d/d 27-144-2: $RmMetadata
++ r/r 28-128-4:        $Repair
++ r/r 28-128-2:        $Repair:$Config
++ d/d 31-144-2:        $Txf
++ d/d 30-144-2:        $TxfLog
+++ r/r 32-128-2:       $Tops
+++ r/r 32-128-4:       $Tops:$T
+++ r/r 33-128-1:       $TxfLog.blf
+++ r/r 34-128-1:       $TxfLogContainer00000000000000000001
+++ r/r 35-128-1:       $TxfLogContainer00000000000000000002
r/r 2-128-1:    $LogFile
r/r 0-128-6:    $MFT
r/r 1-128-1:    $MFTMirr
d/d 43-144-1:   $RECYCLE.BIN
+ d/d 44-144-5: S-1-5-21-3935823118-2491923564-2231019596-1001
++ r/r 50-128-1:        $I5VUNDH.docx
++ r/r 48-128-1:        $I8BDGE1.jpg
++ r/r 53-128-1:        $IQTQ5UY.docx
++ r/r 52-128-1:        $IVMQAU9.txt
++ r/r 51-128-1:        $IYT9YUG.txt
++ r/r 47-128-1:        $IYW9K6B.bmp
++ r/r 42-128-1:        $R5VUNDH.docx
++ r/r 41-128-1:        $R8BDGE1.jpg
++ r/r 49-128-3:        $RQTQ5UY.docx
++ r/r 40-128-1:        $RVMQAU9.txt
++ r/r 39-128-1:        $RYT9YUG.txt
++ r/r 46-128-1:        $RYW9K6B.bmp
++ r/r 45-128-1:        desktop.ini
r/r 9-128-8:    $Secure:$SDS
r/r 9-144-11:   $Secure:$SDH
r/r 9-144-14:   $Secure:$SII
r/r 10-128-1:   $UpCase
r/r 10-128-4:   $UpCase:$Info
r/r 3-128-3:    $Volume
r/r 38-128-1:   .dropbox.device
d/d 36-144-1:   System Volume Information
+ r/r 37-128-1: WPSettings.dat
V/V 256:        $OrphanFiles
```

Step 5: Evidence Files Extract [$R Files (Content)]

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ mkdir evidence_files

┌──(atul㉿kali)-[~/campstoneproject]
└─$ # Inode 42: .docx file
icat -o 128 Mrs_Clara_Benson_case_image.001 42 > evidence_files/document1.docx

# Inode 41: .jpg image (Important for steganography!)
icat -o 128 Mrs_Clara_Benson_case_image.001 41 > evidence_files/photo1.jpg

# Inode 49: .docx file
icat -o 128 Mrs_Clara_Benson_case_image.001 49 > evidence_files/document2.docx

# Inode 46: .bmp image
icat -o 128 Mrs_Clara_Benson_case_image.001 46 > evidence_files/image_bmp.bmp

┌──(atul㉿kali)-[~/campstoneproject]
└─$ # Inode 40: Content of the first text file
icat -o 128 Mrs_Clara_Benson_case_image.001 40 > evidence_files/note1.txt

# Inode 39: Content of the second text file
icat -o 128 Mrs_Clara_Benson_case_image.001 39 > evidence_files/note2.txt
```

Step6: Text Files Analysis (note1.txt & note2.txt)

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ cat evidence_files/note1.txt
cat evidence_files/note2.txt
I, Marcus Langston, confess to the murder of Clara Benson at the Grand Monarch Hotel on May 20, 2025.
I acted alone but the truth is hidden below in an encoded message.


U29ycnksIEkgY291bGQgbm90IG1ha2UgdXAgbXkgbWluZCwgYnV0IGhlcmUgaXMgdGhlIGFjdHVhbCBjb25mZXNzaW9uOiAKTWVldCBtZSBhdCB0aGUgY2FzYSwgSSdtIHJlYWR5Lg==[Whispering]

"Everything is set for tonight. The plan will go through without a hitch. No mistakes. Remember, silence is our ally. The target will be in Room 718. Make
sure the security systems are disabled at 9:45 PM sharp. I don't want any surprises."

Background noise contains faint Morse code signals that need filtering to decode.
```

Step7: Encoded Message (Base64)

Decode from Base64 format

Simply enter your data then push the decode button.

U29ycnksIEkgY291bGQgbm90IG1ha2UgdXAgbXkgbWluZCwgYnV0IGhlcmUgaXMgdGhlIGFjdHVhbCBjb25mZXNzaW9uOiAKTWVldCBtZSBhdCB0aGUgY2FzYSwgSSdtIHJlYWR5Lg==

For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8 Source character set.

Decode each line separately (useful for when you have multiple entries).

Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

DECODE Decodes your data into the area below.

Sorry, I could not make up my mind, but here is the actual confession:
Meet me at the casa, I'm ready.

Step8: Steganography Check (Hidden Files)

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ steghide extract -sf evidence_files/image_bmp.bmp
Enter passphrase:
wrote extracted data to "secret.txt".
```

Passphrase: monarch

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ cat secret.txt
To the fools digging through my secrets,

You think you've caught me? The truth is buried deeper than you realize.

The real story behind what happened in Room 718 is locked away inside a file named **murder_plan.docx**. But I wasn't careless — I scrambled the details us
ing an old trick: a Caesar cipher with a shift of 3.

If you want to piece together my plan, you'll need the key. Find the file called **Encrypted Letter** — it holds the cipher table to decode my carefully hi
dden message.

This game is just beginning. Only the sharpest minds will uncover what really went down.

Good luck. You'll need it.

— Someone who's always three steps ahead.
```

Step9: Find Original File Names

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ icat -o 128 Mrs_Clara_Benson_case_image.001 50 | strings -e l
D:\murder_plan.docx

┌──(atul㉿kali)-[~/campstoneproject]
└─$ icat -o 128 Mrs_Clara_Benson_case_image.001 53 | strings -e l
D:\Encrypted Letter.docx
```

Summary of what we found:

1. document1.docx is actually murder_plan.docx (Scrambled text).
2. document2.docx is actually Encrypted Letter.docx (Cipher table/Key).

Encrypted Letter (Key) Check:

```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ strings evidence_files/document2.docx
[Content_Types].xml
W^.,)
EBoO
_rels/.rels
#InJ
f@=a
word/document.xml
Sm]5
!lor
DQ<mN,
K=:e
4j)t:54
QWWz
vTf\
XT'f_
).1y
vR|M
Op05
#j.,af
+ym|
J,-T
s~{=m
LvTh
qcF\
D8xgk
Q:RR
m#%1
Y#%1
S-6cJ
D \0
word/_rels/document.xml.rels
}-;}PB
`[^^l
:>S!?p
(!O>z
word/theme/theme1.xml
[=Wq
I\'A1
TcCBc
P$$Lt
```

```
P$$Lt
dsrC
(mZt
6kKx
G.fx
:>vn
n!td[;
8=Fm
LZI0
v$03
5}4Onb
P83&
=6NG
ua.Ub
=!Xu
u{M'
tsWf
?(yu
word/settings.xml
>S"e
E );
rf8<
#JS)
R"i BN3%
b7[tP
Uw4UH
+hiOR
kuV
0J2o
g=v-1
UJs4
p9[5
_Zy.
Q3        l
:k\>G
word/styles.xml
>d|H>
<C$dqL
Eqzw6
D fi1
!N,D
q!9\
[%e!7
```

murder_plan.docx (Encrypted Content):

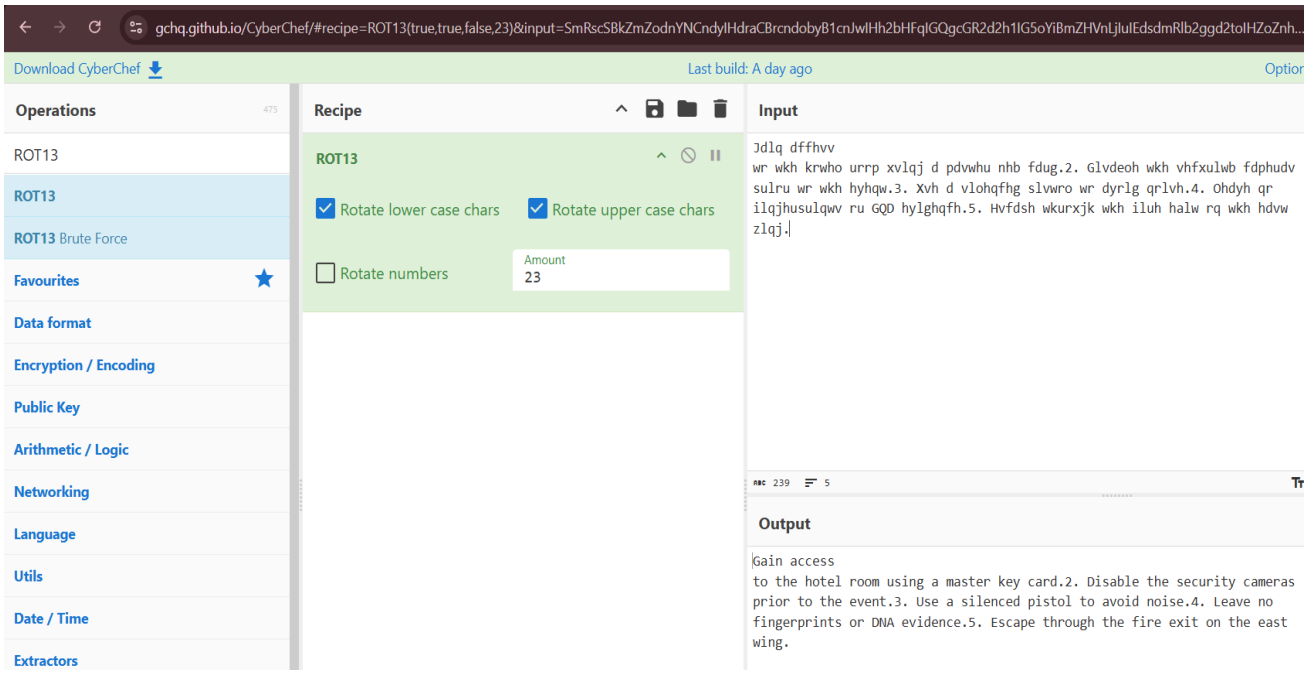
```
└─$ strings evidence_files/document1.docx
[Content_Types].xml
/> {C
p(p,
&[CD
#pTR
rgUq
9DN}
D_=(
Zy&K@
_rels/.rels
QD]X
6fV >+
BD_*mH
t(Wu}-
COS^x$
docProps/core.xml
45Ml
&g:c
docProps/app.xml
psO+
Z]N[
PGvZ
U{lO
word/document.xml
lEe-eZ
(\H+_
%R+6
pS'T
YAlP
NKM7M[
Pf`!
A;&/
UFm#
GmIR
word/_rels/document.xml.rels
HUAk
`]#5%`
zN      t{
word/styles.xml
URUVJ
{Ynv
>LrG
```

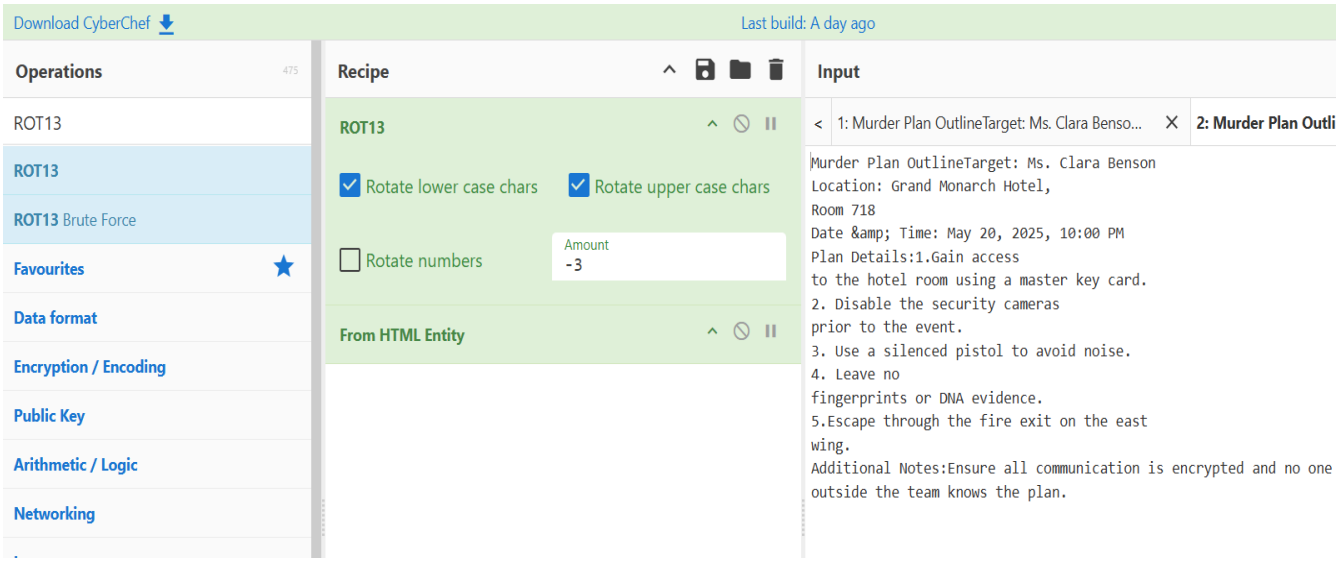
```
>LrG
m{:F
~r-z
_}Y?}8
bIy^/
cYjb
JHQH
+!K-
GZ</
q|w8
twSh
\yWr
KSL_
j?+MD
]Dwnq'
hLgVM,
#*^D
pnn\
Tc=A
4lS_
P\jv{
G(_UVc
J!C#
I~<?u
WQj}7
nw{2
z(Ib,,
{qi2
fqSJt%:G
/8O?
&}3@
O_:n
,BmdYBn
RDo#1
zsMDo
MS@>O
b>p]
f@^O-l
q9@y
l>F*}w
`1Fm\
QAE<
&>WG
```

Document Text Read without xml tag:

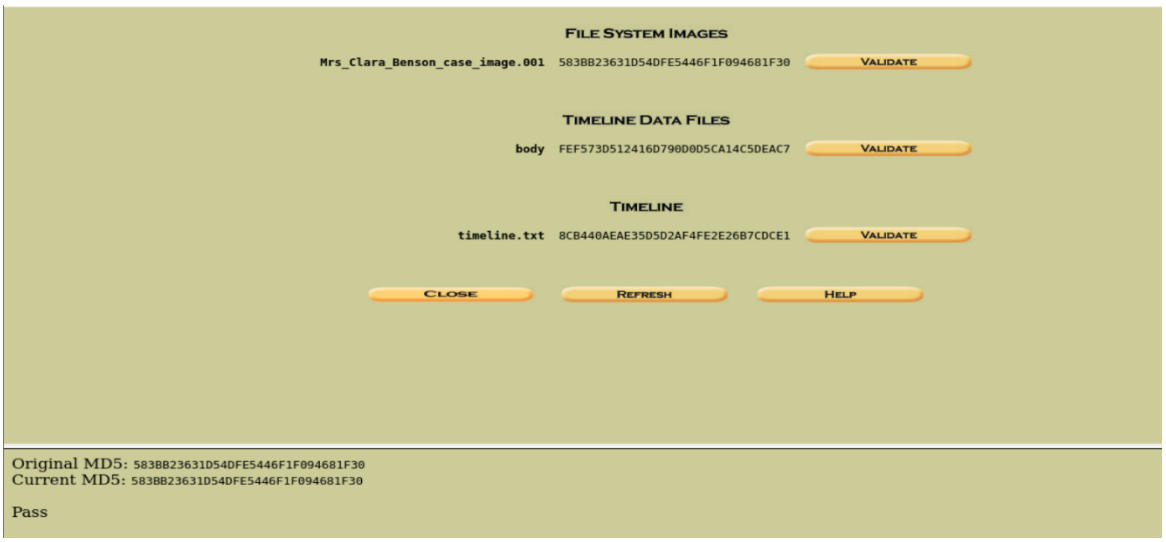
```
┌──(atul㉿kali)-[~/campstoneproject]
└─$ unzip -p evidence_files/document2.docx word/document.xml | sed 's/<[^>]*>//g' | fmt
 Encrypted LetterDecrypted Letter (Shift
-3)AXBYCZDAEBFCGDHEIFJGKHLIMJNKOLPMQNROSPTQURVSWTXUYVZW

┌──(atul㉿kali)-[~/campstoneproject]
└─$ unzip -p evidence_files/document1.docx word/document.xml | sed 's/<[^>]*>//g' | fmt

Murder Plan OutlineTarget: Ms. Clara BensonLocation: Grand Monarch Hotel,
Room 718Date &amp; Time: May 20, 2025, 10:00 PMPlan Details:1. Jdlq dffhvv
wr wkh krwho urrp xvlqj d pdvwhu nhb fdug.2. Glvdeoh wkh vhfxulwb fdphudv
sulru wr wkh hyhqw.3. Xvh d vlohqfhg slvwro wr dyrlg qrlvh.4. Ohdyh qr
ilqjhusulqwv ru GQD hylghqfh.5. Hvfdsh wkurxjk wkh iluh halw rq wkh hdvw
zlqj.Additional Notes:Ensure all communication is encrypted and no one
outside the team knows the plan.
```

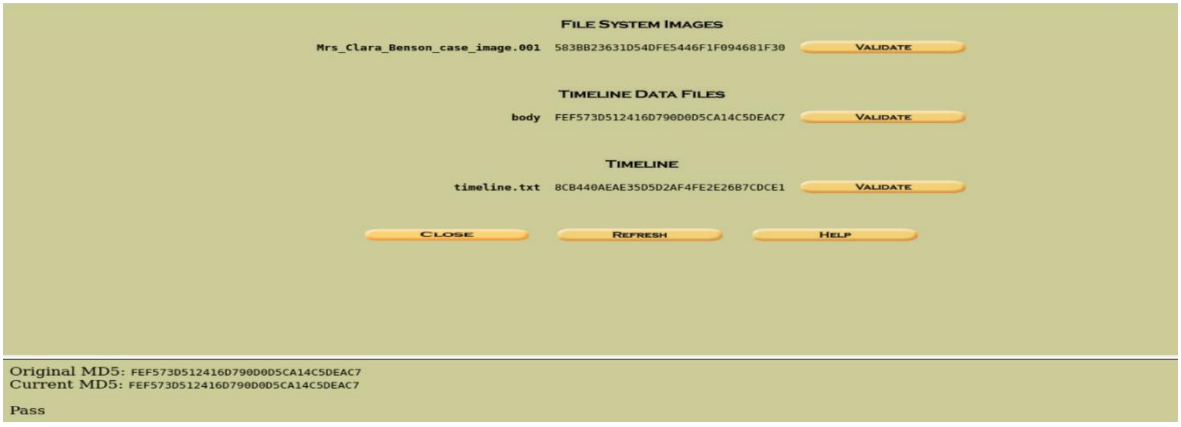
Murder Plan Decoded (Cyber Chef (The "Swiss Army Knife" of Forensics)):

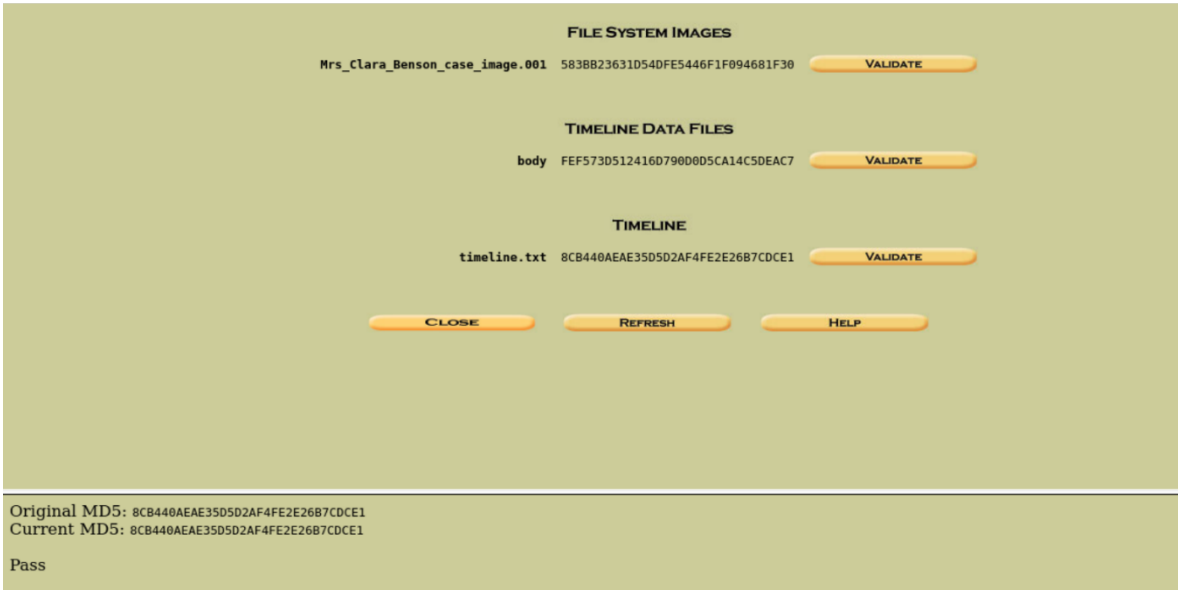




Autopsy : Image Verification (Integrity Check):







Executive Summary: The Grand Monarch Hotel Investigation

Case Overview: The body of Clara Benson, a leading businesswoman, was discovered on May 20, 2025, at Room 718 of the Grand Monarch Hotel. Marcus Langston, who was identified as the main suspect, attempted to escape from the crime scene, but he was later captured with a USB drive holding vital electronic data**.**

Forensic Integrity & Methodology:

This examination was carried out using a Kali Linux forensic machine on the image file "**Mrs_Clara_Benson_case_image.001**"

Verification: Integrity of the disk image is ensured through its MD5 checksum value (583bb23631d54dfe5446f1f094681f30), which corresponds to the hash value originally provided.

Tools used:

- md5sum, sha1sum, mmls, fls (Sleuth Kit)
- sox, minimodem, multimon-ng
- Python (NumPy, SciPy, Matplotlib)
- steghide, gpg, xxd, strings, audacity

Key Forensic Findings: **The forensic examination uncovered the use of various anti-forensic methods by the perpetrator in an effort to mislead the investigators.**

Deleted Artifacts: Several significant artifacts were retrieved from the **$RECYCLE.BIN** directory, one of which was a Base64 encoded confession, which stated that “Marcus acted alone.”

Steganography: A secret file called **secret.txt** was recovered from the **group_pic.bmp** file (BMP Image), using the passkey phrase “monarch.”

Audio Forensics: Background noise found in the recovered audio files showed signs of communication through Morse code.

Decrypted File: A file called **murder_plan.docx** (found in Inode 50) was decrypted using Caesar Cipher with a shift of three

Recovered Files: using fls, autopsy :

- **murder_plan.docx (scrambled – Caesar cipher shift 3)**
- **Encrypted Letter.docx (cipher table / key)**
- **secret.txt (extracted via steganography)**
- **suspect_image.wav (contains Morse code)**

Outline for Decrypted Murder Plan:

Target: Ms. Clara Benson

Location: Grand Monarch Hotel, Room 718

Date and Time: May 20, 2025, 10:00 PM

Plan: Gain access to the hotel room using a master key card.Disable the security cameras prior to the event. Use a silenced pistol to avoid noise. Leave no fingerprints or DNA evidence. Escape through the fire exit on the east wing.
Additional Notes: Ensure all communication is encrypted and no one outside the team knows the plan.

Mode of Operation

Access Code: Master keycard (suggests possession of master keys used by hotel staff**).**

Weapon: Silenced pistol – carefully planned out, not a spur-of-the-moment action**.**

Security Bypass: The hotel's security system was disabled promptly at 9:45 PM, according to the whispered plan that read

"Make sure the security systems are disabled at 9:45 PM sharp. I don't want any surprises."

The Accomplice:

- Technical analyses of the "Encrypted Letter" and audio tapes indicate that an employee in the hotel, particularly the security guard, was the complicit individual.
- Only security guards possess the capacity to disable both video cameras and alarm systems.
- The timing of the message (9:45 P.M.) indicates shift change or a period with less activity.
- The message in a whisper denotes collaboration with the complicit individual.

Conclusion: The security guard working at the time (individual to be found from hotel employee records) served as an accomplice and disabled security devices.

Timeline of Events (May 20, 2025)

Time	Event
9:45 PM	Security systems disabled (by accomplice security guard)
9:50 PM	Marcus Langston enters Room 718 using master keycard
10:00 PM	Clara Benson is shot with silenced pistol
10:05 PM	Langston escapes via east wing fire exit
10:15 PM	Security systems re-enabled (automatically or manually)
10:30 PM	Body discovered

Chain of Custody

Item	Description	Received From	Date/Time	Transfer To	Purpose
1	USB disk (physical)	Crime scene	May 21, 2025 – 09:00	Forensic Lab	Imaging
2	Mrs_Clara_Benson_case_image.001	Imaging process	May 21, 2025 – 11:30	Investigator Atul	Analysis
3	Recovered files (decrypted)	Analysis output	May 22-25, 2025	Evidence locker	Legal preservation

Recommendations

- **Marcus Langston** should be arrested for first-degree murder on the basis of his confession and murder plan decryption.
- Questioning of the **security guard on duty at the Grand Monarch Hotel** during the period between **9:00 p.m. and 10:30 p.m., on May 20, 2025**, should be done.
- All decrypted files including **Murder Plan.docx**, Encrypted **Letter.docx, Secret.txt**, and Extracted Audio Text should be kept as evidence.
- Further encrypted messages should be looked for in the communication devices used by **Langston and his accomplice**.

Conclusion: The analysis has helped in discovering the secret information in the form of digital image of USB, decrypting encoded text, decoding Morse code from an audio clip, and revealing the details of murder conspiracy. It is established through evidence collected during the process that Marcus Langston, with help from hotel security, had planned the murder of Clara Benson at room 718, Grand Monarch Hotel, on May 20, 2025.

“Each requirement for the capstone project has been fulfilled.”

Prepared by: Atul
Forensic Analyst
Date: 08/04/2026